What a Backtest Can Prove

A pre-registered benchmark for portfolio construction, with a power audit and an out-of-sample replication.

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The question

Does a "robust" portfolio method beat a plain one, out of sample, after trading costs, once you account for having tried more than one? The project fixes the decision rule before the run, and then asks a second question that most backtests never ask: **could the rule see the effect it was testing for?**

What was built

- Seven constructions on 18 US ETFs over 168 months (June 2012 to May 2026); five baselines, two treatments, 1,176 rebalances, 10 bps one-way costs.
- A decision rule fixed before the run: a 0.20 net Sharpe bar, a 0.95 probabilistic Sharpe ratio against the best baseline, a 0.05 deflated-Sharpe p-value, and five robustness checks.
- An evaluation apparatus: a leakage suite, a KKT certificate on every solve, and a reproduction check that compares decisions exactly and measurements within a documented bound.

Three findings

1. **The comparator decides.** The same return series passes the deflated Sharpe test against *zero* ($p = 0.0017$) and fails the probabilistic Sharpe test against the best baseline (0.587 against 0.95). With one registered comparison, the multiple-testing correction is inert. What binds is the benchmark.
2. **The gate could not see the bar.** The registered level test required an observed gap of 0.478 to certify a 0.20 effect; its 80%-power threshold was 0.72, and its power at 0.20 was about 24%. On the paired test the estimate is +0.064 [−0.011, +0.147]: a 0.20 advantage is excluded, and the null survives the repair of the test.
3. **The edge did not replicate.** A pre-registered country-ETF sample (18 funds, 252 months, 21 folds) changed the best baseline from sample MVO to equal weight. Equal weight returned 0.3527, `M2` 0.3193, `M1` 0.2710. The benchmark lesson replicated; the treatment edge did not.

`M2` is not shown to be materially better than the best baseline, and a 0.20 advantage is excluded. That is the supported sentence. "No effect" is not.

Method, in one paragraph

Each month, weights are estimated on the previous 60 months and held for one month; the 168 months form 14 non-overlapping 12-month folds. The rule is a conjunction: a 0.20 net Sharpe gap, a 0.95 PSR against the best baseline, a 0.05 deflated-Sharpe p-value, and five robustness checks on turnover, drawdown, cost levels, folds, and concentration. Long-only, 30% cap, 10 bps one-way.

Evidence level

Claim	Status	Why
Single-factor comparison	Exploratory	First sample; the registered test ran without its required blindness
Power audit	Verified	Arithmetic on the published return series
Replication	Confirmatory	Registered before it ran; the new sample did not inform the design
Universal, live, or alpha claim	Not made	One asset family, one period, flat costs, no impact

What it does not claim

- Not a live trading system, a forecast, or a claim of alpha.
- Not a confirmatory test of the first sample: the hypotheses are untested, not falsified.
- Not "no effect": the data are consistent with a small improvement and exclude a material one.
- Not universal: one asset family, one period, current-listed country funds in the replication.

Why it matters

A backtest can be rigorous and still answer the wrong question. The project makes that failure visible, quantifies it, and then runs the same test on a sample that did not inform the design. The reusable output is a discipline: pre-register the rule, audit its power, choose the comparator mechanically, and replicate on fresh territory.

Verify

Repository	github.com/kira-ml/robustfolio
Working paper	essays/robustfolio-working-paper.pdf
Protocol	docs/evaluation_protocol.md
Power correction	notes/paired_inference_2026-11-04.md
Replication	notes/confirmatory_country_etfs_2026-11-05.md

Check the published tables with `python scripts/verify_release.py --archive <your snapshot archive> --run docs/published-run` if you independently hold the snapshot; the archive is not redistributed. Decisions are compared exactly; measurements within 1e-9, with the observed difference printed.

Personal research communication, not investment advice. The project is not a trading system, and no absolute return claim is made.